

HW Trace quick sort as shown in the class for the following array.

i) 85, 76, 99, 13, 45, 103, -5, 26, 31

ii) 350, 790, 400, 200, 100, 96, 900, 890, 760

i j a[i] a[j] a[pivot]

i) 85 76 99 13 45 103 -5 26 31

a[i]

pivot

$a[i] \leq \text{pivot}$

a[j]

$a[j] > \text{pivot}$

85	76	99	13	45	103	-5	26	31
$\bar{x}$	$\bar{x}$	$\bar{i}$						$\bar{j}$

swap $i \& j$

85	76	31	13	45	103	-5	26	99
$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{i}$		$\bar{j}$	$\bar{x}$

swap $i \& j$

85	76	31	13	45	26	-5	103	99
$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{i}$	$\bar{j}$
						$\bar{j}$	$\bar{x}$	

swap pivot $\& j$

-5	76	31	13	45	26	85	103	99
$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{i}$	$\bar{x}$	$\bar{j}$
						$\bar{j}$	$\bar{x}$	

swap pivot $\& j$

-5	26	31	13	45	76	85	103	99
$\bar{x}$	$\bar{i}$	$\bar{j}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$

swap $i \& j$

-5	26	13	31	45	76	85	103	99
$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$
		$\bar{j}$	$\bar{x}$					

swap pivot $\& j$

-5	13	26	31	45	76	85	103	99
-5	13	26	31	45	76	85	$\bar{i}$	$\bar{j}$

swap $i \& j$

-5 13 26 31 45 76 85 99 103

i	j	a[i]	a[j]	a[pivot]
0	8	85	31	85
5	7	103	26	85
7	6	103	-5	85
6	5	85	26	85
1	6	76	85	76
6	5	85	26	76
1	5	85	26	85
2	2	31	31	31
6	5	103	85	85
1	2	26	31	26
7	8	99	103	103

350 790 400 200 100 96 900 890 760

~~x~~ i j ~~x~~ ~~x~~ ~~x~~

swap i & j

350 96 400 200 100 790 900 890 760

~~x~~ ~~x~~ i j ~~x~~ ~~x~~ ~~x~~ ~~x~~

swap i & j

350 96 100 200 400 790 900 890 760

~~x~~ ~~x~~ ~~x~~ ~~x~~ i ~~x~~ ~~x~~ ~~x~~ ~~x~~

swap pivot & j

200 96 100 | 350 | 400 790 900 890 760
 i j
 700 96 200 | 350 | 400 790 900 890 760
 i j
 swap pivot & j swap pivot & j
 96 100 200 350 | 400 790 900 890 760
 96 100 200 350 400 | 790 900 890 760
 swap i & j
 96 100 200 350 400 | 790 760 890 900
 swap pivot & j
 96 100 200 350 400 760 790 890 900
 i j a[i] a[j] a[pivot]
 0 5 350 96 350
 2 4 400 100 350
 4 3 400 200 350
 0 2 200 100 200
 1 0 96 200 200
 0 1 200 96 200
 0 2 100 200 100
 2 1 200 96 100
 4 3 400 760 400
 5 8 790 760 790
 7 6 890 900 790
 7 8 890 900 890